

**STATE OF CALIFORNIA
REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION
895 Aerovista Place, Suite 101
San Luis Obispo, California 93401-7906**

**WASTE DISCHARGE REQUIREMENTS AND MASTER RECYCLING PERMIT
ORDER NO. R3-2011-0217
Waste Discharger Identification No. 3 420104001**

**FOR
LAGUNA COUNTY SANITATION DISTRICT
WASTEWATER RECLAMATION PLANT
PRODUCER OF RECYCLED WATER
SANTA BARBARA COUNTY**

The California Regional Water Quality Control Board, Central Coast Region (hereafter Central Coast Water Board), finds:

PURPOSE OF ORDER

1. The purpose of this Order is to renew and update waste discharge requirements and the master recycling permit for Laguna County Sanitation District (hereafter District, Discharger, or Producer). The District submitted a report of waste discharge on January 27, 2011, for renewed authorization to discharge treated municipal wastewater from the Discharger's Wastewater Reclamation Plant, which serves the unincorporated community of Orcutt, unincorporated Santa Maria, and portions of the City of Santa Maria in Santa Barbara County.
2. California Water Code Section 13510 states that the people of the state have a primary interest in the development of facilities to recycle water containing waste to supplement existing surface and underground water supplies and to assist in meeting the future water requirements of the state.
3. California Water Code Section 13512 states that it is the intention of the legislature that the State undertake steps to encourage development of water recycling facilities so that recycled water may be made available to help meet growing water demands of the State.

FACILITY OWNER AND LOCATION

4. The District owns and operates a tertiary wastewater treatment facility located at 3500 Black Road, Santa Maria (Latitude 34°53'30"N, Longitude 120°30'13"W). The location of the wastewater treatment plant is depicted on Attachment A of this Order.

FACILITY/SITE DESCRIPTION

5. **Service Area** – The District provides wastewater collection, treatment, and disposal of municipal wastewater generated in the unincorporated community of Orcutt, unincorporated Santa Maria, and portions of the City of Santa Maria in Santa Barbara County. The current service area includes approximately 12,000 connections and approximately 128 miles of

collection lines. Wastewater is primarily residential and commercial (domestic in nature) with insignificant industrial contributions.

6. **Water Supply** – Customers within the District's service area use water supplied by Golden State Water Company, the City of Santa Maria, or Foster Water Company, and water supply sources are composed of groundwater and State water with the following characteristics:

Total Dissolved Solids	670 mg/L	Sodium	60 mg/L
Boron	0.13 mg/L	Chloride	66 mg/L
Sulfate	220 mg/L	Nitrate (as NO ₃)	21 mg/L

7. The potable water contains salts and other compounds contributing to hardness. That water exhibits a nuisance quality, as witnessed by the communities' pervasive water softening. Several district-specific studies point to self-regenerating water softeners as a major source of high wastewater salinity. By adding additional salts to the system through the use of water softeners, Laguna County Sanitation District residential and commercial users exacerbate the condition of nuisance salts. Exacerbating the extent of a high salt zone decreases beneficial uses, since salty water has fewer beneficial uses than fresh water. Control of residential self-regenerating water softeners will contribute to the achievement of water quality objectives.
8. **Treatment** - The main treatment system consists of initial screening, primary clarification, trickling filters, secondary clarification, polishing ponds, tertiary filtration, and disinfection using ultraviolet light. A portion of the influent flow (that with highest salts content) is diverted to the salts-reducing treatment system, which consists of a membrane bioreactor with tertiary filtration and reverse osmosis. This flow is then recombined with the main treatment system flow prior to the disinfection process. The treatment plant design capacity is 3.7 million gallons per day (MGD) and current flows average 2.1 MGD. A flow diagram showing the treatment processes is included as Attachment B of this Order. According to the report of waste discharge submitted by the Discharger, the combined wastewater (including tertiary treated and reduced salts portion) has the following characteristics:

Biochemical Oxygen Demand	1.4 mg/L	Sodium	164 mg/L
Suspended Solids	1.4 mg/L	Chloride	169 mg/L
Total Dissolved Solids	707 mg/L	Boron	0.35 mg/L
Sulfate	179 mg/L		

9. **Disposal and Reuse** - Treated municipal wastewater is distributed to a variety of users for landscape irrigation, agricultural purposes, or oil-production processes, and discharged to permanent pasture adjacent to the treatment facilities (used for grazing non-dairy cattle). Active user agreements are in place for reuse of the treated wastewater with Santa Maria Public Airport District; Santa Maria Pacific, LLC; and Punta de la Laguna Properties, LLC. Additional users may enter into agreements with the District with approval of the Central Coast Water Board Executive Officer in accordance with this Order. Approximately 380 MG storage capacity is available for seasonal high flows or when demand for recycled water is low. The disposal and reuse areas are shown on Attachment A of this Order. Brine from the reverse osmosis process is disposed of in a Class 1 non-hazardous injection well regulated by U.S. EPA and not addressed in this Order.

10. **Biosolids Handling and Disposal** – Solids separated from wastewater in the treatment processes are further treated in anaerobic digesters and the resulting biosolids (treated sewage solids) are stored onsite in drying beds and periodically removed and transported to Engel & Gray composting facility for beneficial reuse.

GEOLOGY, SOILS AND WATER

11. **Soils and Geology** – The wastewater reclamation plant is located southwest of the City of Santa Maria and is surrounded by fairly flat agricultural land. Orcutt Creek flows along the west and south sides of the plant. Soils in low-lying areas west of the plant consist of various strata of loose silt, clay, clayey sandy silt, clay sand, sandy clay gravels, cobbles, sand, and pebbles to a depth of 36 feet below ground surface. Various strata of medium-grained sand, black sand, silty clay, clayey coarse sand, sandy clay, and coarse sand exist to a depth of 60 feet below ground surface in low lying areas north of the plant. The soil profile on high ground north of the plant consists of varying layers of silty sand, lean clay, sand with silt, sand, clayey sand, sandy gravel and sandy clay to a depth of 144 feet. Perched water zones in the low-lying areas both west and north of the plant exist that appear to impede or preclude vertical percolation. Areas in high ground north of the plant have shallow sandy clays related to the Orcutt Formation and also appear to perch water.
12. **Surface Water and Groundwater** – Depth to groundwater in the vicinity of the wastewater reclamation plant varies depending upon location and season, and samples indicate groundwater at 14 to 17 feet below ground surface (June and July, 2008). Wet soils, reportedly from perched groundwater, have been measured as shallow as 4.5 feet (1985) below ground surface at the reclamation plant site. Groundwater below the irrigated pasture (disposal area) ranges from 80 to 120 feet below ground surface. The upper effluent holding pond is located approximately 35 feet higher in elevation than the plant, and the lower effluent holding ponds are approximately 40 feet below plant elevation, where the depth to groundwater ranges from 1 to 11 feet below ground surface in the vicinity. Groundwater flows in a northwesterly direction towards the Santa Maria River. Orcutt (Solomon) Creek runs just west and south of the wastewater reclamation plant, as depicted on Attachment A.
13. Mineral analyses (average values from wells on District property) from May 2010 indicate the following groundwater characteristics:

	<u>upgradient wells</u>	<u>downgradient wells</u>
Total Dissolved Solids	826 mg/L	1,873 mg/L
Sodium	186 mg/L	303 mg/L
Chloride	235 mg/L	846 mg/L
Boron	0.3 mg/L	0.4 mg/L
Sulfate	130 mg/L	541 mg/L
Nitrate (as N)	1.3 mg/L	2.5 mg/L

14. **Stormwater** - Federal regulations for stormwater discharges, promulgated by the U.S. Environmental Protection Agency, require specific categories of industrial activities including Publicly Owned Treatment Works (POTWs) to obtain NPDES permits regulating the control of stormwater. The State Water Resources Control Board has adopted general NPDES permits for stormwater discharges associated with industrial facilities. Stormwater at the District's wastewater plant is collected from process areas and treated as wastewater and disposed of

as recycled water. Therefore, stormwater is regulated with this Order and a separate stormwater permit is not required.

BASIN PLAN

15. The Central Coast Water Board has adopted the *Water Quality Control Plan, Central Coast Basin* (the Basin Plan), which designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for receiving waters within the Region.

16. **Surface Water Beneficial Uses** – Present and anticipated beneficial uses of Orcutt Creek include:

Municipal and Domestic Supply	Agricultural Supply	Cold Fresh Water Habitat
Water Contact Recreation	Wildlife Habitat	Groundwater Recharge
Non-Contact Water Recreation	Estuarine Habitat	Freshwater Replenishment
Commercial and Sport Fishing	Rare, Threatened, or Endangered Species	

This Order does not authorize discharge to surface waters; however, protection of beneficial uses is important as the discharges may have direct and indirect impacts to surface waters.

17. **Groundwater Beneficial Uses** – Present and anticipated beneficial uses of groundwater in the vicinity of the wastewater reclamation plant include Municipal, Domestic, Agricultural, Industrial Process, and Industrial Service Supply.

18. The Basin Plan specifies median water quality objectives for certain groundwater basins, which are intended to serve as a baseline for evaluating water quality management in the basin. The objectives are, at best, representative of gross areas only. The discharge area is located on the border between the Orcutt Sub-area and the Upper Guadalupe Sub-area of the Santa Maria Sub Basin. Corresponding median groundwater quality objectives specified in the Basin Plan are as follows:

	<u>Orcutt Sub-area</u>	<u>Upper Guadalupe Sub-area</u>
Total Dissolved Solids	740 mg/L	1000 mg/L
Sodium	65 mg/L	230 mg/L
Chloride	65 mg/L	165 mg/L
Boron	0.1 mg/L	0.5 mg/L
Sulfate	300 mg/L	500 mg/L
Nitrate (as N)	2.3 mg/L	1.4 mg/L

19. **TMDL** - Total maximum daily load (TMDL) allocations will be developed for impaired surface waters in the Central Coast Region. TMDL documents will allocate responsibility for constituent loading throughout the watershed. If TMDLs determine constituent contributions from waste discharged may adversely impact beneficial uses or exceed narrative or numeric water quality objectives, changes in these waste discharge requirements may be required. Waste discharge requirements may be modified to implement applicable TMDL provisions and recommendations. TMDL listings for the Santa Maria River include fecal coliform, *E. coli.*, toxicity, pesticides, sodium, chloride, unionized ammonia, and nutrients. Water Board staff is currently developing Santa Maria River TMDLs.

RECYCLED WATER

20. **Recycled Water** – Title 22, Division 4, Chapter 3 of the California Code of Regulations specifies State Department of Public Health (DPH) criteria for use of recycled water. Water Code section 13523 authorizes the Regional Board to issue reclamation requirements for water that is used as reclaimed (recycled) water. The Central Coast Water Board has consulted with the State and County Health Departments regarding these reuse requirements.
21. **Master Reclamation Permit** – Pursuant to section 13523.1 of the Water Code, The Central Coast Water Board may, in lieu of issuing waste discharge requirements pursuant to Section 13263 or water reclamation requirements pursuant to Section 13523 for a user of reclaimed water, issue a master reclamation permit to a supplier or distributor, or both, of reclaimed water. These master reclamation permits must, at a minimum, include the adoption of waste discharge requirements, requirements consistent with the uniform statewide reclamation criteria pursuant to Section 13521, requirements to establish enforceable user requirements, and require quarterly reporting and periodic inspections of facilities using reclaimed water. This Order also serves as a master reclamation permit pursuant to Section 13523.1 of the Water Code.
22. **Recycled Water Policy** - The Strategic Plan Update 2008-2012 for the Water Boards includes a priority to increase sustainable local water supplies available for meeting existing and future beneficial uses by 1,725,000 acre-feet per year, in excess of 2002 levels, by 2015, and ensure adequate water flows for fish and wildlife habitat. The State Water Resources Control Board (State Water Board) adopted the Recycled Water Policy (Resolution No. 2009-0011) on February 3, 2009. The Recycled Water Policy is intended to support the Strategic Plan priority to promote sustainable local water supplies. Increasing the acceptance and promoting the use of recycled water is a means towards achieving sustainable local water supplies. The Recycled Water Policy is also intended to encourage beneficial use of, rather than solely disposal of, recycled water.
23. The Recycled Water Policy calls for the development of regional groundwater basin/sub-basin salt/nutrient management plans. The State Water Board recognizes that, pursuant to the letter from statewide water and wastewater entities dated December 19, 2008, and attached to Resolution No. 2009-0011 adopting the Policy, the local water and wastewater entities, together with local salt/nutrient contributing stakeholders, will fund locally driven and controlled, collaborative processes open to all stakeholders that will prepare salt and nutrient management plans for each basin/sub-basin in California, including compliance with CEQA and participation by Central Coast Water Board staff.
24. It is the intent of the Recycled Water Policy that salts and nutrients from all sources be managed on a basin-wide or watershed-wide basis in a manner that ensures attainment of water quality objectives and protection of beneficial uses. The appropriate way to address salt and nutrient issues is through the development of regional or sub-regional salt and nutrient management plans rather than through imposing requirements solely on individual projects. The Central Coast Water Board finds that a combination of regional management plans and individual or programmatic project requirements may be necessary to protect beneficial uses. One of the primary components of the required regional salt/nutrient management plans is the development and implementation of groundwater basin/sub-basin

monitoring programs. As specified in the Recycled Water Policy, salt/nutrient contributing stakeholders will be responsible for conducting, compiling, and reporting the monitoring data once the regional groundwater monitoring programs are developed.

ANTIDEGRADATION

25. State Water Board Resolution No. 68-16 (*Statement of Policy with Respect to Maintaining High Quality of Waters in California*) requires Regional Water Boards, in regulating the discharge of waste, to maintain high quality waters of the State unless it is demonstrated that any change in quality will be consistent with the maximum benefit to the people of the State, will not unreasonably affect beneficial uses, and will not result in water quality less than that described in a Regional Water Board's policies (i.e., quality that exceeds applicable water quality standards). Resolution No. 68-16 also states, in part:

Any activity which produces or may produce a waste or increased volume or concentration of waste and which discharges or proposes to discharge to existing high quality waters will be required to meet waste discharge requirements which will result in best practicable treatment and control of the discharge necessary to assure that (a) a pollution or nuisance will not occur and (b) the highest water quality consistent with maximum benefit to the people of the State will be maintained.

The discharge regulated by this Order is subject to waste discharge requirements that will result in treatment, control, prevention of pollution and nuisance, and maintenance of water quality consistent with maximum benefit to the people of the State. As such, these waste discharge requirements are consistent with the provisions of Resolution No. 68-16.

MONITORING PROGRAM

26. Monitoring and Reporting Program (MRP) No. R3-2011-0217 is part of this Order. The MRP requires routine wastewater influent, effluent and receiving water (groundwater) sampling and analysis to verify compliance with this Order. Monitoring reports are required monthly and an annual report is required by January 30th of each year. The MRP is included as Attachment C.

CALIFORNIA ENVIRONMENTAL QUALITY ACT (CEQA)

27. These waste discharge requirements are for an existing facility and therefore are exempt from provisions of the CEQA in accordance with California Water Code Section 15301.

EXISTING ORDERS AND RESOLUTIONS

28. The discharge is currently regulated by Waste Discharge Requirements and Master Recycling Permit Order No. 01-042, adopted by the Central Coast Water Board on May 18, 2001.
29. **Statewide General Waste Discharge Requirements for Sanitary Sewer Systems (Collection System WDR).** – Collection System WDR, Order No. 2006-0003-DWQ, was adopted May 2, 2006, and applies to publicly owned sanitary sewer systems (collection

systems) that are one mile or greater in length. The Collection System WDR requires collection system entities to develop a Sanitary Sewer Management Plan (SSMP). These SSMPs are required to include goals; organization; legal authority; operations and maintenance program; design and performance provisions; an overflow emergency response plan; fats, oils, and greases control program; systems evaluations and capacity assurance program; monitoring, measures, and program modifications; and an SSMP Program audit. Additionally, the Collection System WDR requires the collection system entities to report sanitary sewer overflows (SSOs). Collection system entities are required to report SSOs that are greater than 1,000 gallons. Furthermore, some entities must also report SSOs less than 1,000 gallons discharging to surface waters or storm drains or that threaten public health. Reporting provisions are set forth in the Collection System WDR. Reporting occurs through the Statewide Online SSO database. The Discharger is enrolled under the Collection System WDR and implements its provisions.

GENERAL FINDINGS

30. No discharge of waste to waters of the State creates a vested right to continue the discharge. All discharges of waste into waters of the State are privileges, not rights. A permit is conditional upon the discharge complying with provisions of Division 7 of the California Water Code and of the Clean Water Act (as amended or as supplemented by implementing guidelines and regulations) and requirements necessary to implement water quality control plans, protect beneficial uses, and prevent nuisance. Compliance with this Order should ensure that water quality is protected.
31. This Order contains restrictions on individual pollutants. The effluent limitations for biochemical oxygen demand (BOD) and total suspended solids (TSS) are based on achievable limits for secondary treatment as demonstrated by historical facility effluent data. Effluent limitations in this Order for total dissolved solids, sodium and chloride have been scientifically derived to implement water quality objectives that protect beneficial uses. Both the beneficial uses and the water quality objectives have been approved pursuant to state law. All beneficial uses and water quality objectives contained in the Basin Plan were approved under state law and submitted to and approved by U.S. Environmental Protection Agency (EPA) prior to May 30, 2000. The requirements of this Order take into consideration past, present, and probable future beneficial uses of the receiving waters, the environmental characteristics, including water quality of the Santa Maria River hydrographic unit, and coordinated control of all factors which affect water quality in the area.
32. On September 19, 2011, the Central Coast Water Board notified the Discharger and interested agencies and persons of its intent to consider adoption of waste discharge requirements for the discharge and provided them with a copy of the proposed Order and an opportunity to submit written comments and scheduled a public hearing. Written comments were required to be received by October 20, 2011.
33. In a public hearing on **December 1, 2011**, the Central Coast Water Board heard and considered all comments pertaining to the discharge, all evidence in the record, and the applicable law and found this Order consistent with the above findings.
34. Any person aggrieved by this action of the Water Board may petition the State Water Board to review the action in accordance with Water Code section 13320 and California Code of

Regulations, title 23, sections 2050 and following. The State Water Board must receive the petition by 5:00 p.m., within 30 days of the adoption date of this Order, except that if the thirtieth day following the date of the order falls on a Saturday, Sunday, or state holiday, the petition must be received by 5:00 p.m. on the next business day. Copies of the law and regulations applicable to filing petitions may be found on the internet at http://www.waterboards.ca.gov/public_notices/petitions/water_quality or will be provided upon request.

IT IS HEREBY ORDERED that, pursuant to authority in the California Water Code, Division 7, including Sections 13263, 13267, and 13523, Laguna County Sanitation District, its agents, successors, and shall comply with the following:

All technical and monitoring reports submitted pursuant to this Order are required pursuant to Section 13267 of the California Water Code. Failure to submit reports in accordance with schedules established by this Order or attachments to this Order, or failure to submit a report of sufficient technical quality to be acceptable to the Executive Officer, may subject the Discharger to enforcement action pursuant to Section 13268 of the California Water Code.

General conditions, definitions and the method of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated January 1984, and referenced in paragraph H.3 of this Order.

Throughout these requirements footnotes are listed to indicate the source of requirements specified. Requirement footnotes are as follows:

CWC = California Water Code
BP = Basin Plan
T22 = California Code of Regulations, Title 22, Recycled Water Criteria
DPH = State Department of Public Health

Requirements without footnotes are based on staff's professional judgment.

A. DISCHARGE PROHIBITIONS

1. Discharge to areas other than the disposal facilities shown on Attachment A of this Order or reuse sites approved by the Executive Officer is prohibited.^{T22, CWC}
2. Discharge of any wastes including overflow, bypass, seepage, overspray and runoff from transport, treatment, or disposal systems to adjacent properties, adjacent drainage ways, or to waterways is prohibited.^{T22, CWC}
3. Discharge of untreated or partially treated wastewater is prohibited.^{CWC}
4. Discharge of wastewater within 100 feet of any well used for domestic supply is prohibited.^{T22} Discharge of recycled water within 50 feet of a water supply wells is prohibited.^{DPH} All impoundment of disinfected tertiary recycled water shall comply with setback requirements set forth in Section 60310 of the California Code of Regulations.^{T22}

5. The treatment, storage, distribution, or reuse of recycled water in a manner that creates a nuisance as defined in section 13050(m) of the California Water Code is prohibited.^{CWC}
6. Cross-connections are prohibited between potable water supply and pipes containing recycled water. Supplementing recycled water with water used for domestic supply shall not be allowed except through an air-gap separation, which complies with the requirements of Section 7602(a) and 7603(b) of Title 17, California Code of Regulations (CCR).^{DPH}
7. Transportation of undisinfected recycled water within a pipeline used to transport disinfected tertiary treated recycled water is prohibited.^{DPH}
8. Use of recycled water for direct human consumption or for processing of food or drink intended for human consumption is prohibited.^{DPH}
9. Pond freeboard less than two (2) feet is prohibited unless the pond is specifically designed for a different freeboard. All ponds shall be protected from erosion, washout and flooding from a rainfall event having a predicted frequency of once in 100 years.
10. Discharge of radioactive substances is prohibited.^{BP}

B. DISCHARGE SPECIFICATIONS

1. The annual average effluent shall not exceed 3.7 MGD.
2. Effluent discharged to the District's irrigated land disposal area shall not exceed the following:

Table 1: Land Disposal Effluent Limitations

Constituent	Units	Monthly Average	Daily Maximum
Settleable Solids	mL/L	0.1	0.4
BOD, 5-Day	mg/L	30	90
Suspended Solids	mg/L	30	90
Total Dissolved Solids*	mg/L	900	
Sodium*	mg/L	180	
Chloride*	mg/L	180	
pH	within the range 6.5 to 8.4 ^{BP}		
Dissolved Oxygen	at all times not less than 2.0 mg/L		

mL/L – milliliters per liter

mg/L – milligrams per liter

* Compliance based upon 12-month running mean.

C. RECYCLED WATER SPECIFICATIONS

Note - Following reuse requirements apply in addition to effluent limitations specified above.

1. Discharger has developed, and shall maintain with updated information, an Engineering Report on the Production, Distribution and Use of Recycled Water (Engineering Report) in conformance with Title 22 of the California Code of Regulations, for review and approval of the

Executive Officer (after consultation with State and local health departments). Updates to the Engineering Report must be submitted no less than six months in advance of proposed significant changes to treatment processes.

2. Recycled water production and use shall at all times be in conformance with recycled water criteria established in Title 22, Division 4, Chapter 3 of the California Code of Regulations and the Engineering Report^{T22, CWC}. Recycled water shall be adequately oxidized, coagulated, clarified, filtered, disinfected^{T22} and not exceed the following limitations:

Table 2: Recycled Water Limitations

Constituent	Units	Monthly Average (30-day)	Daily Maximum
Settleable Solids	mL/L	--	0.1
Suspended Solids	mg/L	10	25
Sulfate	mg/L	300	
Boron	mg/L	0.5	

3. The median number of coliform organisms in recycled water shall not exceed 2.2 MPN per 100 mL, as determined from the bacteriological results of the last seven days for which analyses have been completed. The number of coliform organisms shall not exceed 23 MPN per 100 mL in more than one sample in any 30-day period and shall not exceed 240 MPN per 100 mL in any single sample.^{T22}
4. Turbidity of the filtered recycled water shall not exceed 2 NTU within a 24-hour period, 5 NTU more than 5 percent of the time within a 24-hour period, and 10 NTU at any time.^{T22}
5. Delivery of reclaimed water for landscape irrigation and agricultural uses shall cease as soon as possible and all wastewater shall be returned to the treatment and/or disposal system if:
 - a. Disinfection of wastewater ceases at any time; or,
 - b. Recycled water specifications are violated or threaten to be violated.
6. Recycled water shall be confined to the authorized reuse areas identified in the engineering report or recycled water user agreements.
7. Recycled water shall not be used for irrigation during extended periods of rainfall and/or runoff.
8. Personnel involved in producing, transporting or using recycled water; or those involved in inspecting, maintaining or operating any distribution system equipment for recycled water shall be informed of possible health hazards that may result from contact and use of recycled water.^{T22}
9. Use of recycled water shall occur at a time and in a manner to prevent or minimize public contact with recycled water and to prevent ponding in irrigation areas.
10. Areas irrigated with recycled water shall be posted in English and Spanish to warn that recycled water is being used. Signs shall be no less than four inches high by eight inches

wide and include the wording "RECYCLED WATER – DO NOT DRINK."^{T22}

11. Recycled water use areas shall be properly labeled and regularly inspected to ensure proper operation, absence of leaks, and absence of illegal connections.^{T22} Recycled water valves shall be of a design to prevent public access.
12. Drinking fountains shall be protected from recycled water spray, mist or runoff.
13. Tank trucks used to transport recycled water shall be appropriately labeled and shall not leak.
14. Except as allowed under section 7604 of title 17, California Code of Regulations, no physical connection shall be made or allowed to exist between any recycled water system and any separate system conveying potable water.^{T22}
15. The portions of the recycled water piping system that are in areas subject to access by the general public shall not include any hose bibs. Only quick couplers that differ from those used on the potable water system shall be used on the portions of the recycled water piping system in areas subject to public access.^{T22}
16. The Producer shall implement, and ensure that users implement annual employee training to ensure proper operation of reclamation facilities, worker protection, and compliance with this Order.
17. Prior to providing recycled water to potential users, the District must enter into agreement with the user and obtain approval from the Central Coast Water Board. Requests for approval must include the following information:
 - a. Name of responsible party,
 - b. Location of recycled water application,
 - c. Plans and specifications for recycled water distribution systems,
 - d. Proof of employee/user training,
 - e. Proof of cross-connection tests,
 - f. Inspection schedule,
 - g. Proof that the user is familiar and agrees to comply with the recycled water use requirements in this Order,
 - h. Quantity of water used,
 - i. Method of recycled water use,
 - j. Storage facilities,
 - k. Confirmation of signs posted,
 - l. Producer/User agreement.

D. RECEIVING WATER (GROUNDWATER) LIMITATIONS

(Receiving water quality is a result of many factors, some unrelated to the discharge. This permit considers these factors and is designed to minimize the influence of the discharge to receiving waters.)

1. The discharge shall not cause groundwater to contain taste- or odor-producing substances in concentrations that adversely affect beneficial uses.^{BP}
2. The discharge shall not cause radionuclides to be present in concentrations that are deleterious to human, plant, animal, or aquatic life or result in the accumulation of radionuclides in the food web to an extent which presents a hazard to human, plant, animal, or aquatic life.^{BP}
3. The discharge shall not cause groundwater to contain concentrations of organic or inorganic chemicals in excess of the limiting concentrations set forth in California Code of Regulations, Title 22, Division 4, Chapter 15, Article 5.5, Section 64444 (organic) and Article 4, Section 64431 (inorganic).^{BP}
4. The discharge shall not cause groundwater to contain concentrations of chemical constituents in amounts that adversely affect the agricultural supply beneficial use. Interpretation of adverse effects shall be as described in University of California Agricultural Extension Service guidelines provided in Table 3-3 of the Central Coast Basin Pla.^{BP}
5. The discharge shall not cause a significant increase in mineral constituent concentrations in the underlying groundwater, as determined by comparison of samples collected from wells located upgradient and downgradient of the disposal area.^{BP}
6. The discharge shall not cause underlying groundwater to contain concentrations of constituents in excess of water quality objectives listed in Finding No. 17.

E. PRETREATMENT SPECIFICATIONS

1. The Discharger is exempt from applicable pretreatment requirements specified under 40 CFR 125.66(d). In accordance with requirements specified in this Order, the Discharger shall implement public education and waste minimization/source reduction programs to limit the introduction of toxic pollutants and pesticides into the treatment plant. Implementation of a pollution prevention program will substitute for those requirements specified under 40 CFR 125.66 (d) (Nonindustrial Source Control Program).

F. SALT AND NUTRIENT MANAGEMENT PROGRAM

1. The Discharger shall maintain an ongoing salt/nutrient management program with the intent of reducing mass loading of salts and nutrients (with an emphasis on nitrogen species) in treated effluent to a level that will ensure compliance with effluent limitations and protect beneficial uses of groundwater.
2. Salt reduction measures shall focus on all potential salt contributors to the collection system, including water supply, commercial, industrial and residential dischargers. The salt/nutrient management program shall also address the concentration of salts in the wastewater treatment process as a result of excessive hydraulic retention times and/or chemical addition.
3. Nutrient reduction measures shall focus on optimizing wastewater treatment processes for nitrification and denitrification, or other means of nitrogen removal. Reduction measures may

also include source control (non-human waste from commercial and industrial sources) as appropriate.

4. As part of the salt/nutrient management program, the Discharger shall submit an annual report of salt and nutrient reduction efforts. This salt/nutrient management report shall be included as part of the annual report described in Monitoring and Reporting Program No. R3-2011-0217. The report shall be submitted by January 30th, and shall include (at a minimum):

Salt Component

- a. Calculations of annual salt mass discharged to (influent) and from (effluent) the wastewater treatment or recycling facility with a description of contributing sources;
- b. Analysis of wastewater evaporation/salt concentration effects;
- c. Analysis of groundwater monitoring results for salts constituents and associated trends;
- d. Analysis of potential impacts of salt loading on the groundwater basin (focusing on the relationship between salt concentration in the discharge and the Basin Plan water quality objectives);
- e. Summary of existing salt reduction measures; and,
- f. Recommendations and time schedules for implementation of any additional salt reduction measures.

Nutrient Component

- a. Calculations of annual nitrogen mass (for all identified species) discharged to (influent) and from (effluent) the wastewater treatment or recycling facility with a description of contributing sources;
 - b. Analysis of wastewater treatment facility ability to facilitate nitrification and denitrification, or other means of nitrogen removal;
 - c. Analysis of groundwater monitoring results for nitrogen constituents and trends;
 - d. Analysis of potential impacts of nitrogen loading on the groundwater basin (focusing on the relationship between salt concentration in the discharge and the Basin Plan water quality objectives);
 - e. Summary of existing nitrogen loading reduction measures; and,
 - f. Recommendations and time schedules for implementation of any additional nitrogen loading reduction measures.
5. As an alternative to the salt/nutrient management program requirements described above, upon Executive Officer approval, the Discharger may submit documentation and summary of participation in a regional salt/nutrient management plan implemented under the provisions of State Water Board Resolution No. 2009-0011 (Recycled Water Policy).

G. BIOSOLIDS SPECIFICATIONS

Biosolids refers to non-hazardous sewage sludge as defined in 40 CFR 503.9. Sewage sludge that is hazardous (as defined in 40 CFR 261) must be disposed of in accordance with requirements of the Resource Conservation Recovery Act (RCRA). Sludge with PCB levels in excess of 50 mg/kg must be disposed in accordance with 40 CFR 761.

1. All biosolids generated by the Discharger shall be used or disposed of in compliance with the applicable portions of the following regulations.
 - a. 40 CFR 503 - for biosolids that are land applied, placed in surface disposal sites (dedicated land disposal sites or monofills), or incinerated.
 - b. 40 CFR 258 - for biosolids disposed of in municipal solid waste landfills.
 - c. 40 CFR 257 - for all biosolids use and disposal practices not covered under 40 CFR 258 or 503).
 - d. 40 CFR 503 Subpart B (land application) applies to biosolids applied for the purpose of enhancing plant growth or for land reclamation. Section 503 Subpart C (surface disposal) applies to biosolids placed on the land for the purpose of disposal.

The Discharger is responsible for ensuring that all biosolids produced at its facility are used or disposed of in accordance with these rules, whether the Discharger uses or disposes of the biosolids itself or transfers them to another party for further treatment, use, or disposal.

H. PROVISIONS

1. Order No. 01-0042, Waste Discharge Requirements and Master Recycling Permit for Laguna County Sanitation District, adopted by the Central Coast Water Board on May 18, 2001, is hereby rescinded, except for enforcement purposes.
2. Discharger shall comply with Monitoring and Reporting Program No. R3-2011-0217 (included as Attachment C of this Order), as ordered by the Executive Officer.
3. Discharger shall comply with all items of the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated January 1984 (included as Attachment D of this Order).
4. This Order may be reopened to address any changes in State or Federal plans, policies, or regulations that would affect requirements for the discharge.
5. Discharger shall maintain a comprehensive operation and maintenance manual for the wastewater treatment, storage, and disposal facilities. Discharger shall also develop, update as needed, and maintain as part of its operations and maintenance manual a mercury handling plan. The mercury handling plan shall (at a minimum) address inventory, inspection, monitoring, incident response, worker safety and training.

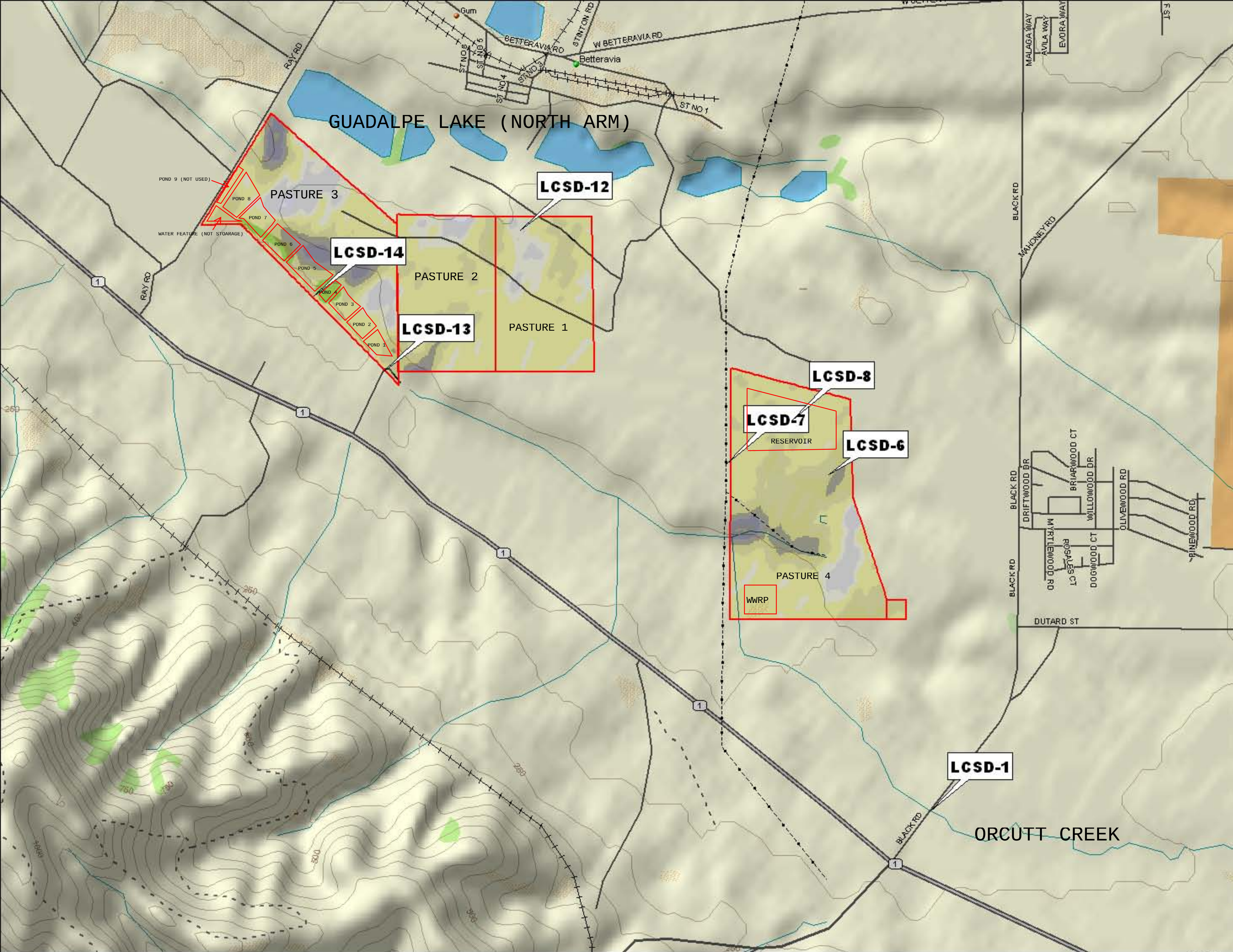
6. Treatment, storage, and disposal facilities shall be managed to exclude the public and posted to warn the public of the presence of wastewater.
7. Pursuant to California Code of Regulations Title 23, Chapter 3, Subchapter 9, the Discharger shall submit a Report of Waste Discharge to the Executive Officer not later than **June 1, 2015**, addressing: a) Whether there will be changes in the continuity, character, location or volume of the discharge; and, b) Whether, in their opinion, there is any portion of the Order that is incorrect, obsolete or otherwise in need of revision.

I, Roger W. Briggs, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an order adopted by the California Regional Water Quality Control Board, Central Coast Region on December 1, 2011.

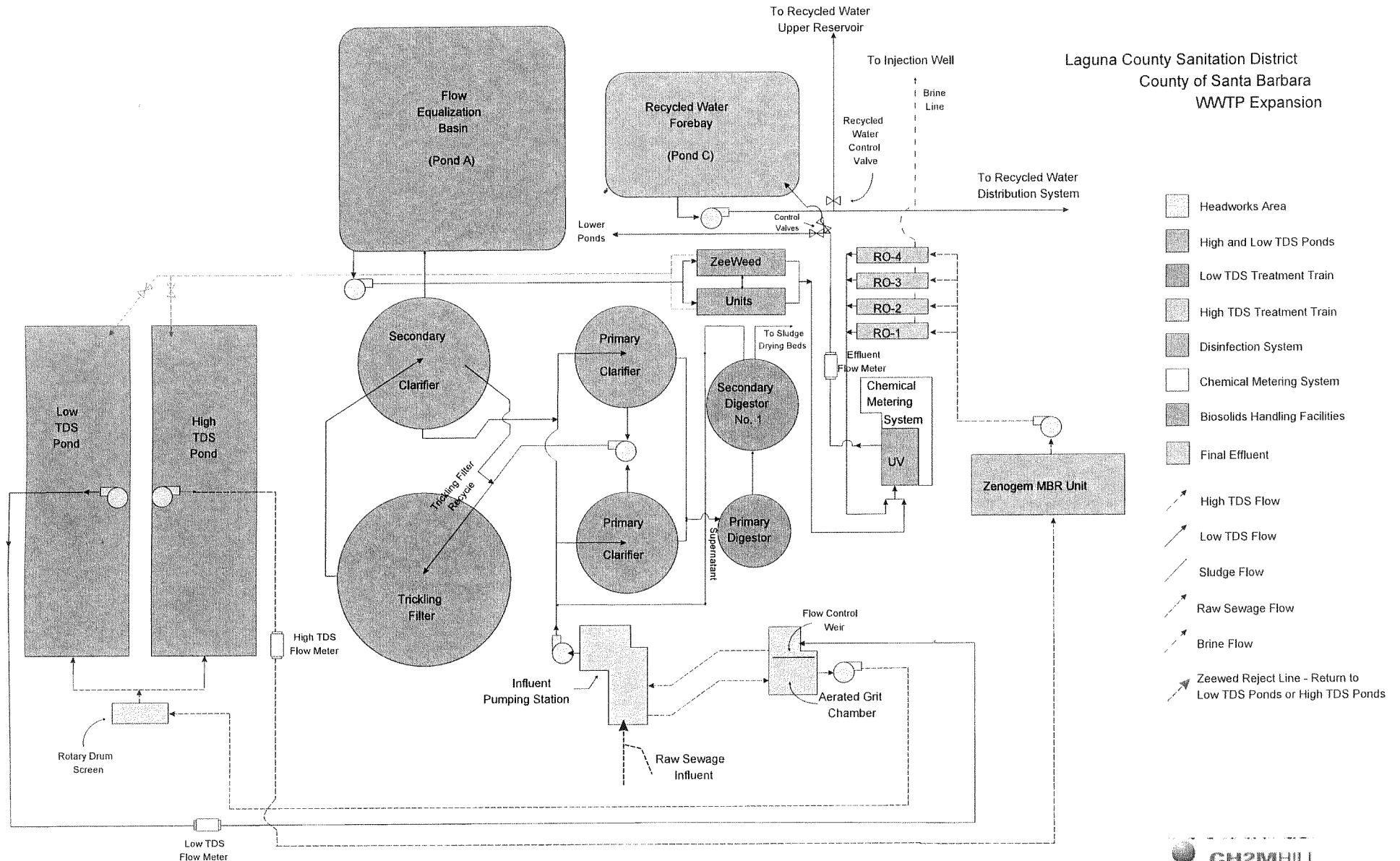


Executive Officer

Attachment A – Location and Service Area
Attachment B – Treatment Process Diagram
Attachment C – Monitoring and Reporting Program
Attachment D – Standard Provisions and Reporting Requirements



Laguna County Sanitation District
County of Santa Barbara
WWTP Expansion



- Headworks Area
- High and Low TDS Ponds
- Low TDS Treatment Train
- High TDS Treatment Train
- Disinfection System
- Chemical Metering System
- Biosolids Handling Facilities
- Final Effluent
- High TDS Flow
- Low TDS Flow
- Sludge Flow
- Raw Sewage Flow
- Brine Flow
- ZeeWeed Reject Line - Return to Low TDS Ponds or High TDS Ponds

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION**

**MONITORING AND REPORTING PROGRAM NO. R3-2011-0217
FOR
LAGUNA COUNTY SANITATION DISTRICT
WASTEWATER RECLAMATION PLANT
(PRODUCER & USER OF RECYCLED WATER)
SANTA BARBARA COUNTY**

Reporting responsibilities are specified in Sections 13225(a), 13267(b), 13383, and 13387(b) of the California Water Code. This discharge monitoring program is issued in accordance with Provision G.2 of Central Coast Water Board Order No. R3-2011-0217.

Influent Monitoring

Representative samples of the wastewater reclamation plant influent shall be collected and analyzed as follows:

Constituent	Units	Type of Sample	Sampling & Analysis Frequency
Flow Volume	MGD	Metered	Daily
Maximum Daily Flow	MGD	Metered	Monthly
Mean Daily Flow	MGD	Calculated	Monthly
Biochemical Oxygen Demand (5-day)	mg/L	24-hr composite	Monthly
Total Suspended Solids	mg/L	24-hr composite	Monthly
Total Nitrogen (as N) (all forms identified)	mg/L	24-hr composite	Quarterly

Effluent Monitoring

Representative samples of the effluent discharge shall be collected and analyzed as follows:

Constituent	Units	Type of Sample	Sampling & Analysis Frequency
Flow Volume to each discharge/reuse location	MG	Estimate	Daily
Turbidity ¹	NTU	Metered	Continuously
Total Coliform Bacteria ¹	MPN/100 mL	Grab	Daily
Settleable solids	mL/L	Grab	Weekly
Biochemical Oxygen Demand (5-day)	mg/L	24-hr composite	Weekly
Total Suspended solids	mg/L	24-hr composite	Weekly
pH	pH units	Grab	Weekly
Freeboard in all ponds	Feet	Measure	Weekly
Total Dissolved Solids	mg/L	24-hr composite	Quarterly
Sodium	mg/L	24-hr composite	Quarterly
Chloride	mg/L	24-hr composite	Quarterly
Sulfate	mg/L	Grab	Quarterly
Total Nitrogen (as N) (all forms identified)	mg/L	Grab	Quarterly

Constituent	Units	Type of Sample	Sampling & Analysis Frequency
Aluminum	mg/L	24-hour composite	Annually
Arsenic	mg/L	24-hour composite	Annually
Beryllium	mg/L	24-hour composite	Annually
Boron	mg/L	24-hour composite	Annually
Cadmium	mg/L	24-hour composite	Annually
Chromium, Total	mg/L	24-hour composite	Annually
Cobalt	mg/L	24-hour composite	Annually
Copper	mg/L	24-hour composite	Annually
Flouride	mg/L	24-hour composite	Annually
Iron	mg/L	24-hour composite	Annually
Lead	mg/L	24-hour composite	Annually
Lithium	mg/L	24-hour composite	Annually
Manganese	mg/L	24-hour composite	Annually
Mercury	mg/L	24-hour composite	Annually
Molybdenum	mg/L	24-hour composite	Annually
Nickel	mg/L	24-hour composite	Annually
Selenium	mg/L	24-hour composite	Annually
Vanadium	mg/L	24-hour composite	Annually
Zinc	mg/L	24-hour composite	Annually
CEC ²	mg/L	24-hour composite	Annually

¹ Effluent monitoring for turbidity and coliform bacteria applies only to water supplied for reuse.

² Constituents of Emerging Concern (CEC) shall include endocrine disrupters, personal care products or pharmaceuticals identified by the State Water Board pursuant to the State Recycled Water Policy.

Groundwater Monitoring

Representative samples of groundwater from wells located up-gradient and down-gradient from the discharge area shall be monitored for the following constituents:

Parameter	Units	Type of Sample	Sampling & Analysis Frequency
Depth to Groundwater	Feet	Measured	Annually (April)
Total Dissolved Solids	mg/L	Grab	Annually (April)
Sodium	mg/L	Grab	Annually (April)
Chloride	mg/L	Grab	Annually (April)
Sulfate	mg/L	Grab	Annually (April)
Boron	mg/L	Grab	Annually (April)
Total Nitrogen (as N) (all forms identified)	mg/L	Grab	Annually (April)

At the Discharger's request, the Central Coast Water Board Executive Officer may approve participation in a basin-wide salt/nutrients management plan implemented under the provisions of State Water Board Resolution No. 2009-0011 (Recycled Water Policy) in lieu of the groundwater monitoring described above.

Biosolids Monitoring

The following information shall be submitted with the Annual Report (due January 30th). If no biosolids are removed from the facility during the reporting period (year), then the Discharger shall include such statement in the Annual Report.

- a. Volume of biosolids removed and disposal and/or reuse destination. Order or permit number (if applicable) for the biosolids destination shall also be provided.
- b. Representative sample of biosolids removed from the drying beds shall be analyzed for the following parameters: Arsenic, Cadmium, Copper, Lead, Mercury, Molybdenum, Nickel, Selenium, Zinc, Total Nitrogen, and % solids.
- c. Biosolids shall be identified as Class A or Class B (in accordance with criteria specified at 40CFR 503). The basis for classification shall also be described.
- d. Pathogen reduction and vector attraction reduction achievement methods shall be described in adequate detail to demonstrate compliance with 40CFR 503.32.

Effluent Salt & Nutrient Management Program Reporting

The Discharger shall report the status of its efforts to reduce effluent salts and nutrients in its Annual Report. The status reports shall include detailed descriptions of all measures implemented by the Discharger according to Salt & Nutrient Management Program requirements specified in section F of this Order.

Reporting

Monthly self-monitoring reports shall be submitted by the last day of the month following the monitoring period, and shall summarize the results of all monitoring performed during that period. The Discharger shall electronically submit Self-Monitoring Reports (SMRs) using the State Water Board's California Integrated Water Quality System (CIWQS) Program Web site (<http://www.waterboards.ca.gov/ciwqs/index.html>). In the event there will be service interruption for electronic submittal, the Discharger shall submit self-monitoring reports electronically to centralcoast@waterboards.ca.gov. Additionally, an Annual Report shall be submitted by January 30th and shall include:

- a. Tabular and graphical summaries of the monitoring data obtained during the preceding year. Duplicate copies of monthly reports are not necessary and do not fulfill requirements for "summaries".
- b. Discussion of any and all incidents of non-compliance and corrective actions taken to ensure compliance is restored.
- c. List of facility staff and corresponding certification levels.
- d. Summary of biosolids monitoring, as described above.
- e. Summary of collection system management plans, or reference report submitted under separate cover as required by this or separate sanitary sewer requirements.
- f. Summary of salts and nutrient management program implementation.

- g. Summary of recycled water distribution and documentation of User compliance.



ORDERED BY: _____
Roger W. Briggs, Executive Officer

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